

RAIL RADAR

Live Indian Railway Intelligence & Dynamic Machine Learning ETA Forecasting

Fusing real-time GPS telemetry, route timetables, and gradient-boosted decision trees to predict arrival shifts.

The Core Innovation

- 12-Feature Gradient Boosting Regressor
- Dynamic arrival delay shifts per halt
- Replaces naive static extrapolations
- Sub-5ms machine learning inference
- Confidence-scored arrival estimates

Situational Intelligence

- 60/40 Operational Split Screen layout
- Interactive Leaflet map with pulsing train
- Real-time network KPIs & delay alerts
- 8,490 Indian Railways trains searchable
- Space-efficient station timeline auto-scroll

The Hackathon Team

Anuj Maurya

Frontend / Map & Dashboard

Archita Kesharwani

Backend / Real-Time Engine

Aru Shubham Singh

ML / ETA Prediction

Shlok Bhardwaj

Data Analytics & Research

Live Prototype Status: Production Ready (Backend on Render | Frontend on Vercel | 59 Passing Tests)

Tech Stack: FastAPI (Python 3.11+) | React 19 | Vite 8 | Tailwind CSS 4 | Scikit-learn | Leaflet | Joblib

Public GitHub Repository: <https://github.com/anuj-maurya-01/RailRadar>

Traditional Train Tracking Fails 24 Million Commuters Daily

Static timetables and crude single-checkpoint extrapolations generate severe information asymmetry.

01. Static Timetable Fiction

Traditional rail tracking services assume trains will magically stick to scheduled timetables despite operational disruptions.

KEY LIMITATIONS:

- Ignores cumulative network congestion
- Cannot foresee junction bottlenecks
- Blinds passengers to snowball delays
- Leads to missed platform connections
- Platform overcrowding & panic

Result: High passenger anxiety

02. Naive Extrapolation

Existing tracking applications copy-paste the latest reported station delay across all remaining stops without adjustment.

FLAWED LOGIC:

- If +20 min at Stop 3, assumes exactly
- +20 min at Stop 25 (false assumption)
- Misses track speed recovery zones
- Overlooks rush-hour congestion dips
- Fails to model single-track waits

Result: Wildly inaccurate ETAs

03. Massive National Scale

Indian Railways operates 13,000+ trains daily across 68,000+ km of tracks, connecting 7,300+ stations.

SYSTEMIC IMPACT:

- 24M+ daily commuters impacted
- Station platform overcrowding
- Operational friction for crew schedulers
- Lack of forward-looking predictions
- Zero delay confidence metrics

Solution: Machine Learning Engine

Rail Radar: Live Indian Railway Intelligence

Replacing guesswork with machine-learning arrival predictions and real-time situational awareness.

Dynamic ML Delay & ETA Forecasting

- Pre-trained HistGradientBoostingRegressor analyzes 12 operational features.
- Differentiates between recorded ground delay and predicted corridor delay.
- Dynamically calculates expected arrival and departure times per upcoming stop.
- Sub-5ms inference enables real-time updates without database slowdowns.

60/40 Operational Split Dashboard

- 60% Interactive Leaflet Map: Polyline route corridors and station halos.
- Custom SVG Locomotive Marker with real-time pulsing radar beacon.
- 40% Selected Train Telemetry: Current speed, delay badges, and ML deltas.
- Zero third-party mapping API keys required; OpenStreetMap tiles powered.

Network KPIs & Critical Delay Alerts

- 4 Clickable Operational KPI Cards: Active Trains, On Time, Delayed, Cancelled.
- Real-time automated alert thresholding ($\geq 10m$ Warning, $\geq 25m$ Alert).
- Station Operations Table with in-table search, sorting, and pagination.
- Compact station timeline with auto-scroll to active halt & 'Jump to Live'.

Enterprise Resilience & Zero Downtime

- Graceful Fallback Engine: Recovers on HTTP 429 upstream rate limit errors.
- Bundled 8,490 Indian Railways train database indexed in-memory.
- Mathematical coordinate interpolation fills missing GPS station waypoints.
- 59 Automated Unit and Integration Tests guarantee operational stability.

Modern Three-Tier Architecture & Data Pipeline

Decoupled, high-throughput asynchronous proxy with client security and rapid inference.

1. Frontend Client Tier

REACT 19 * VITE 8 * TAILWIND 4

- 60/40 Split Operational UI layout
- TrainMap (Leaflet hardware accelerated)
- Custom SVG Locomotive Pulsing Marker
- SelectedTrainPanel live telemetry
- Bounded RouteTimeline with auto-scroll
- Instant debounced search across 8,490 trains
- Non-blocking 30s background sync polling
- Dual Light/Dark transportation themes

Hosting: Vercel Cloud Edge

HTTPS * Client SPA * Vercel CDN

2. API Gateway & Logic

FASTAPI * UVICORN * PYTHON 3.11+

- GET /api/trains/live-summary
- GET /api/trains/search-suggestions
- GET /api/train/{train_number}
- GET /api/health (Uptime Monitor)
- Strict CORS & API Key isolation
- Data Normalizer & Schema Validator
- Mathematical Station Interpolation
- 12-Feature Engineering Pipeline
- Graceful HTTP 429 rate-limit failover

Hosting: Render Web Service

render.yaml Blueprint * Health Check

3. ML & Data Fabric

SCIKIT-LEARN * JOBLIB * PANDAS

- HistGradientBoostingRegressor
- Pre-fitted OneHotEncoder (Categoricals)
- Sub-5ms inference per train query
- Dual Data Feed Architecture:
 - 1. Live RailRadar GPS Telemetry
 - 2. Bundled 8,490-Train Database
- APScheduler Background Harvester
- JSONL Telemetry Recording
- Dynamic arrival delta calculators

Offline Fallback Database:

8,490 Pre-indexed Route Schedules

12-Feature HistGradientBoosting Delay Forecaster

Statistical delay adjustments accounting for non-linear corridor dynamics, speed, and time.

Engineered Input Feature Matrix (12 Variables)

Feature Name	Type	Operational Significance
current_delay_min	Numeric	Baseline ground delay at latest halt
distance_covered_km	Numeric	Track progress from route origin
distance_remaining_km	Numeric	Remaining distance to final terminus
scheduled_travel_time_min	Numeric	Total timetable allocated duration
time_elapsed_min	Numeric	Actual journey operational time elapsed
current_speed_kmh	Numeric	Instantaneous GPS/segment velocity
average_speed_kmh	Numeric	Cumulative running velocity with halts
station_progress_pct	Numeric	Route completion fraction (0.0 - 1.0)
day_of_week	Categoric	Monday - Sunday (One-Hot Encoded)
time_of_day	Categoric	Morning, Afternoon, Evening, Night
rush_hour	Categoric	Peak congestion corridor flag (0 or 1)
train_type	Categoric	Service priority (RAJ, SHT, SF, EXP)

Mathematical Formulation & Reasoning

DYNAMIC ETA FORMULA:

ETA = Timetable Arrival + Delay_live + y_ML

where y_ML = f(Speed, Distance, RushHour, TrainType)

Why HistGradientBoostingRegressor?

- Native Binning: Handles continuous spatial & speed features without lag.
- Non-Linear Dynamics: Captures how delays compound during rush hours.
- Sub-5ms Inference: Essential for real-time live dashboard sync.
- Outlier Immunity: Prevents single GPS glitches from skewing predictions.
- Missing Data Tolerance: Seamlessly processes partial station reports.

Verified by automated test suite (59 test cases passing)

60/40 Operational Split Transportation UI

Engineered for cognitive ease, rapid triage, and high-density telemetry scanning.

60% Map Canvas

HARDWARE ACCELERATED LEAFLET

- Full-canvas route polyline
- Pulsing SVG Locomotive marker
 - displaying real-time heading & speed
- Waypoint halos distinguishing passed stops from upcoming halts
- Dynamic bounding box centering on train selection
- High-contrast OpenStreetMap tiles
 - (Zero third-party API keys required)

Visual Focus:

Instant corridor orientation without clutter.

40% Telemetry Panel

SELECTED TRAIN INTELLIGENCE

- Selected Train telemetry card
- Color-coded delay pill:
 - (Green <=5m, Yellow <=20m, Red >20m)
- Model Delay vs. Live Delay comparison
- Next station distance & ETA countdown
- Origin-Destination Corridor badge
- Space-efficient vertical Timeline:
 - Auto-scrolls to active train halt
 - 'Jump to Live' 1-click focus

Space Efficiency:

Bounded container avoids page bloat.

Multi-Train Operations

TRIAGE, SEARCH & CONTROLS

- Smart Autocomplete Search bar:
 - Debounced across 8,490+ trains
- 4 Real-time KPI Metric cards:
 - Clickable filtering (All / On Time / Delayed)
- High-Severity Delay Alerts Banner:
 - Instant notification of 15m+ disruptions
- Station Operations Table:
 - In-table filter, column sort, pagination
- Non-blocking 30s live telemetry sync

Theme Engine:

Modern Light & High-Tech Dark modes.

Engineered for High Concurrency & Fault Tolerance

Resilient architectural patterns ensuring uninterrupted operation under real-world network constraints.

Graceful HTTP 429 Rate-Limit Fallback

- Live railway APIs frequently impose strict query quotas.
- Rail Radar detects HTTP 429 status codes automatically.
- Instantly fails over to the bundled 8,490-train offline dataset.
- End-users experience zero application crashes or blank states.

Mathematical Coordinate Interpolation

- Many smaller rural stations lack explicit GPS coordinate records.
- Algorithm finds nearest upstream and downstream known junctions.
- Computes distance-weighted latitude/longitude coordinates.
- Guarantees contiguous polyline route mapping without missing stops.

Zero-Leak Credential Isolation

- API tokens and Bearer credentials remain strictly on the FastAPI server.
- The React client communicates exclusively with the local/cloud proxy.
- Restrictive CORS headers block unauthorized cross-site requests.
- Telemetry logs redact sensitive headers before JSONL persistence.

Automated Test Verification (59 Tests)

- 59 Comprehensive automated unit and integration tests.
- Covers normalization, 12-feature vectors, ML inference, & rate limits.
- Full test execution time: < 2.0 seconds.
- Validates frontend production build output via npm run build.

Measurable Hackathon Achievements

From concept to a production-grade, deployed intelligence platform in record time.

8,490+

Trains Indexed

National coverage with instant autocomplete.

< 5ms

ML Inference Latency

Sub-second turnaround for real-time updates.

59 / 59

Tests Passing

100% automated test pass rate on backend.

100%

Cloud Deployed

Render Backend + Vercel Edge Frontend.

What We Accomplished During the Hackathon:

1. Built and production-tuned the 12-feature HistGradientBoosting delay prediction pipeline.
2. Designed an end-to-end 60/40 operational dashboard using React 19, Tailwind 4, and Leaflet.
3. Engineered real-time network KPI cards, automatic delay alerts, and interactive route mapping.
4. Solved upstream API rate-limiting via seamless in-memory indexing of 8,490 train schedules.
5. Automated deployment workflows for Render Blueprint and Vercel Edge hosting.
6. Authored extensive production documentation, API schemas, and an open-source MIT repository.

Future Roadmap: Scaling Rail Radar Nationally

Expanding from predictive passenger tracking to a comprehensive transit intelligence network.

Phase 1: Real-Time Feeds

MONTHS 1 - 3

- **WebSocket Streaming:**
Sub-second live locomotive push updates replacing 30s polling.
- **Doppler Weather Radar:**
Incorporate monsoon rain and dense winter fog intensity into ETA model.
- **Mobile PWA Companion:**
Offline-first mobile web app with push notifications for station halts.
- **Multi-Lingual Support:**
Dashboard localization in 10+ languages.

Phase 2: Network Effects

MONTHS 3 - 6

- **Secondary Delay Cascades:**
Graph neural networks modeling rake turnarounds & track ripple effects.
- **Passenger Crowd Telemetry:**
Zero-permission Bluetooth beacon & onboard GPS ground-truth verification.
- **Station Platform Allocation:**
Predictive platform arrival estimates preventing platform bottlenecking.
- **Automated Delay Explanations:**
Natural language root cause summaries.

Phase 3: Controller Suite

MONTHS 6+

- **Dispatcher Decision Support:**
Assisting section controllers with track precedence simulations.
- **Smart Junction Precedence:**
Reinforcement learning algorithms for prioritizing fast trains at chokepoints.
- **Energy Optimization:**
Speed curve recommendations to reduce traction power waste during idle halts.
- **Nationwide Fleet Analytics:**
Zone-by-zone operational scorecards.

Predictable Railway Travel for Every Commuter

Rail Radar brings transparency, predictive precision, and peace of mind to Indian Railways.

The Hackathon Team

Anuj Maurya

Frontend / Map & Dashboard

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Role: HistGradientBoosting, 12 Features

Shlok Bhardwaj

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Role: Delay Analytics & KPI Modeling

Repository & Open Source

Public GitHub Repository:

<https://github.com/anuj-maurya-01/RailRadar>

- License: MIT Open Source License
- Live Backend: Render Cloud Service (FastAPI)
- Live Frontend: Vercel Cloud Edge (React 19)
- Test Coverage: 59 Automated Tests Passing

Thank You!

We are excited to answer your questions.

Rail Radar: Live Indian Railway Intelligence

Making train travel predictable and stress-free.